

MKID simulation through
automated geometry
generationMKID Design
Automation

Microwave kinetic inductance detectors (MKIDs) are cryogenic superconducting photon detectors that can achieve time, space and energy resolution.

An MKID consists of an inductor-capacitor resonator whose resonant frequency can be tuned by varying its inductance and capacitance.

Traditional MKID design methods relied on manually creating and simulating a small number of designs, and interpolating between those points.

This poster presents a user-friendly program that automatically generates designs by allowing the user to vary the parameters highlighted here.

It is crucial in automation that the developer converts *implicit* understandings that human users might have into *explicit* instructions to allow the system to operate.

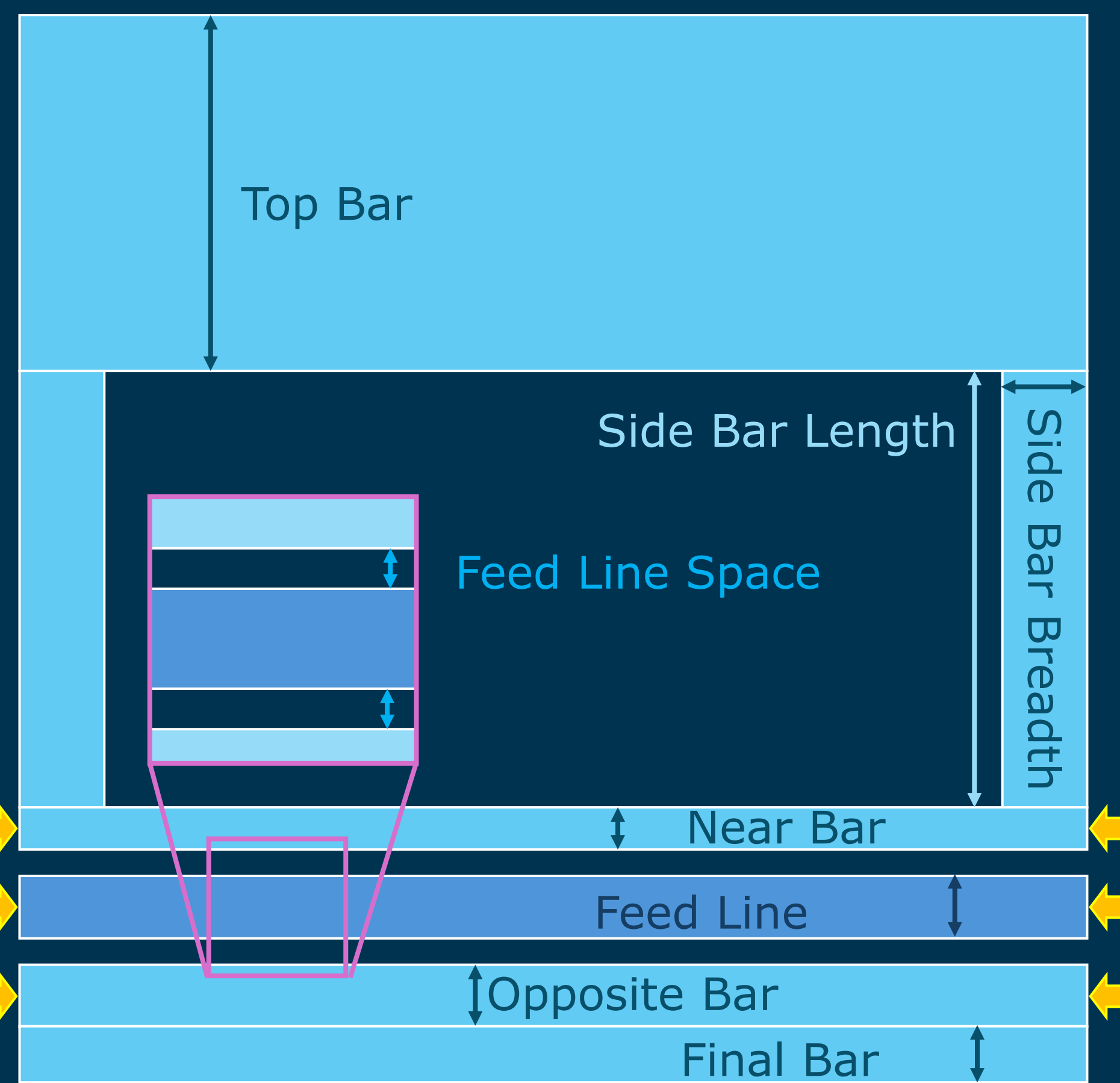
We have focussed on structuring simulations into physics-informed classes to bundle logical groupings of circuit elements. The methods of these classes generate repeating structures and manage the interaction between related and dependent parameters.

Ground Plane

The **Ground Plane** is constructed using its tab which sets the breadth and spacing parameters of the **Bars** and **Feed Line**.

Ports are placed at the midpoints of bars they are attached to.

Origin of **Capacitor** offset from inner dimensions of **Ground Plane** by a spacing parameter, while the length of the side bar is calculated from the y -size of the Capacitor and Inductor.



GUI Features

The GUI is coded using Tkinter and allows the user to set the dimensions of the MKID. Each element has a tab and a set of SpinBoxes with step-sizes determined by the cell size.

The interface procedurally generates and then plots the MKID layout as shown

The user can save .son Sonnet geometry files or the visual representation of the resonator using the buttons on the GUI.

Upcoming features include iterating over dimensions to create multiple MKID designs in a single pass and direct Sonnet simulation interface.

[github.com/creanero/
mkid_sonnet_variation](https://github.com/creanero/mkid_sonnet_variation)

References: [1] Sonnet Software Inc (2018) "Sonnet User's Guide", Release 16
[2] McAleer, Cáthail, et al. (2024) "Automation of MKID Simulations for Array Building with AEM (Automated Electromagnetic MKID Simulations)", Journal of Low Temperature Physics
[3] Creaner, Oisín, et al. (2026) "Simulation of MKID Design Sensitivity to Fabrication Tolerances", IEEE Transactions on Applied Superconductivity

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Artificial Intelligence tools (Claude, Copilot) were used to support code implementation based on human-generated design specifications and code bases.

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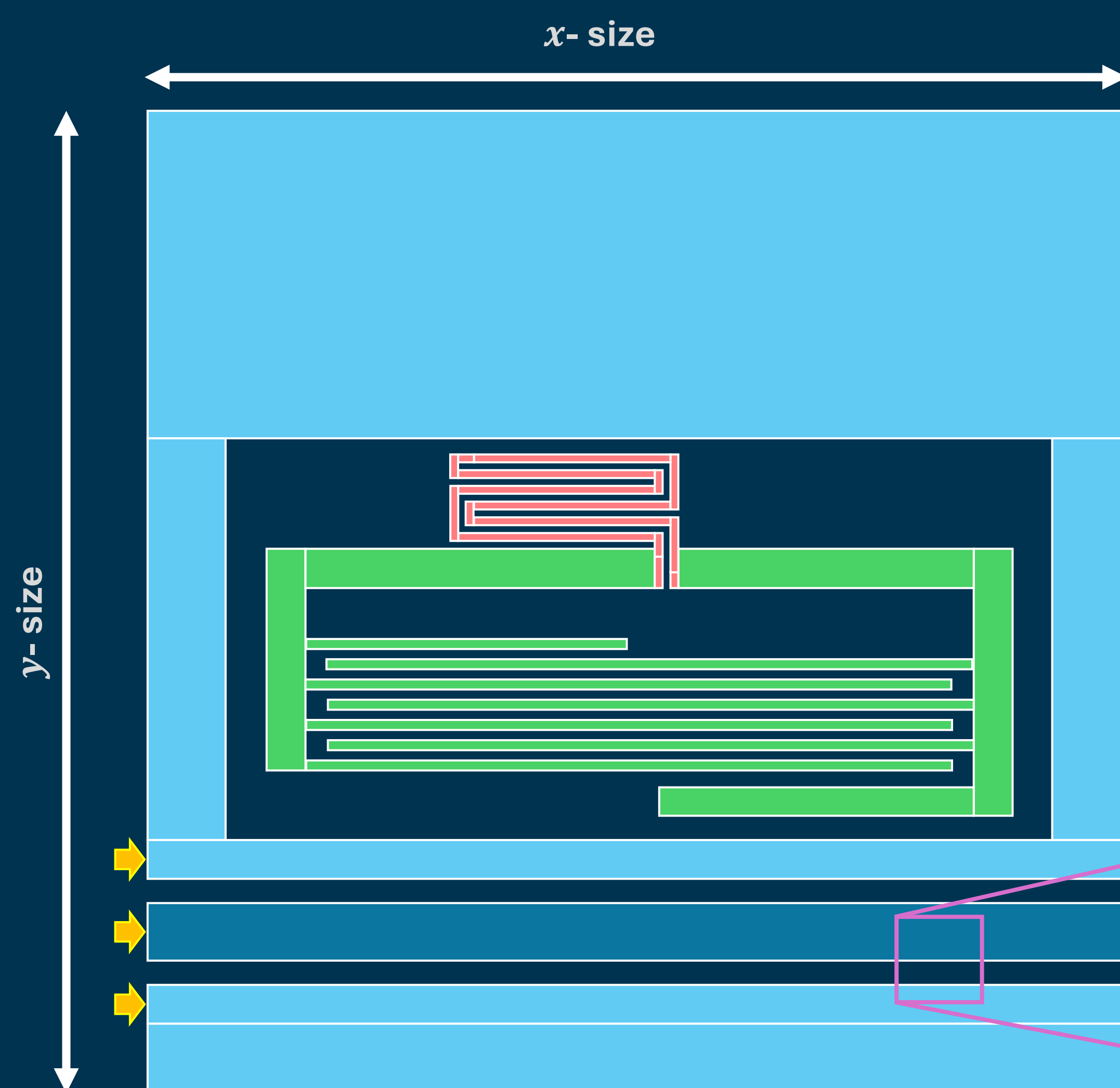


Figure 1: Schematic representation of an MKID. The circuit consists of a ground plane and feed line in blue, an inductor in red and a capacitor in green. The ports are illustrated in yellow.

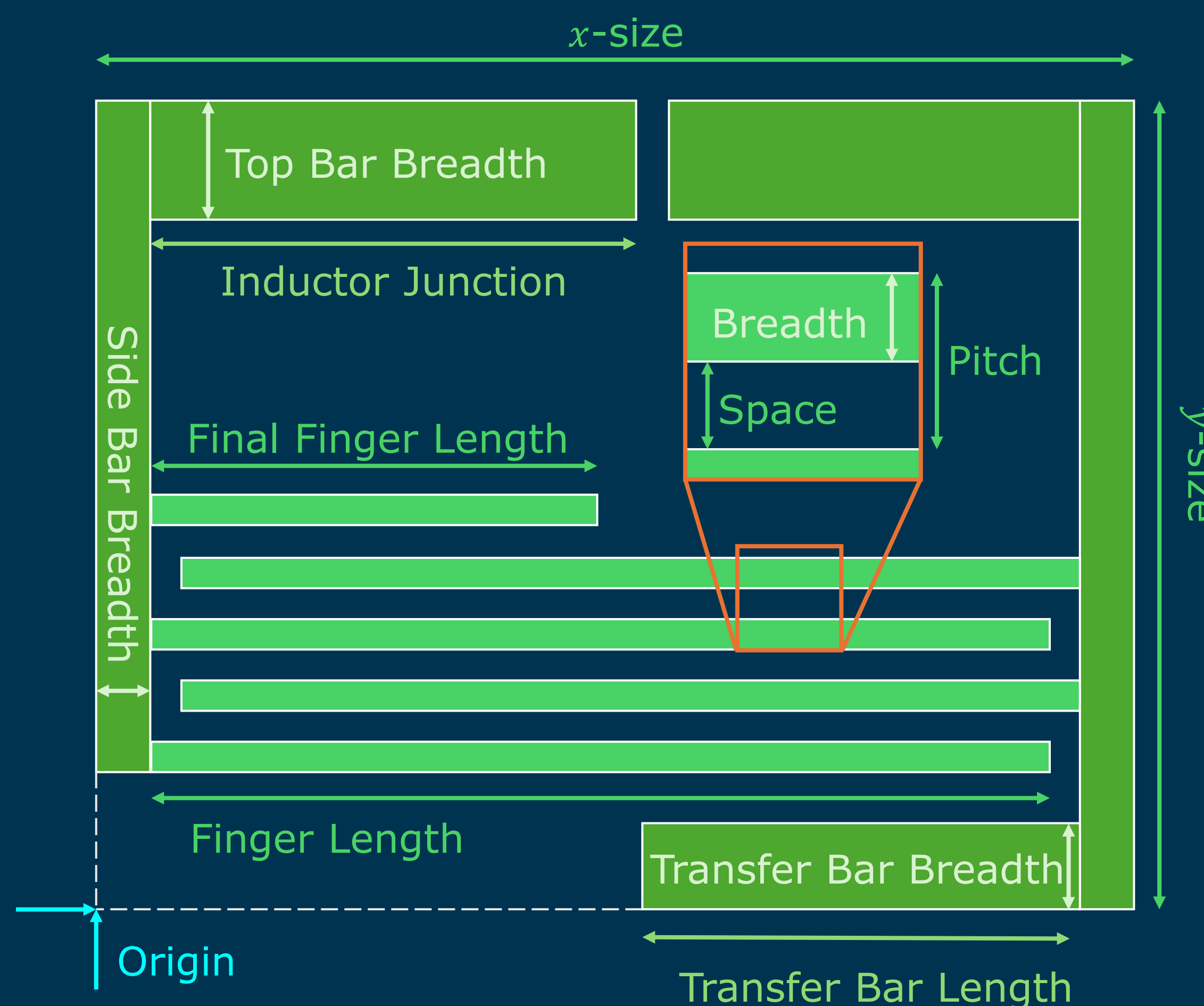
Capacitor

The **Capacitor** is constructed of a **Frame** and **Fingers**.

The **Frame** method sets the length and breadth of bars it's made of.

The **Finger** method sets their Breadth, Length, Pitch and Count.

Origin of **Inductor** calculated from the Junction in the **Capacitor**.



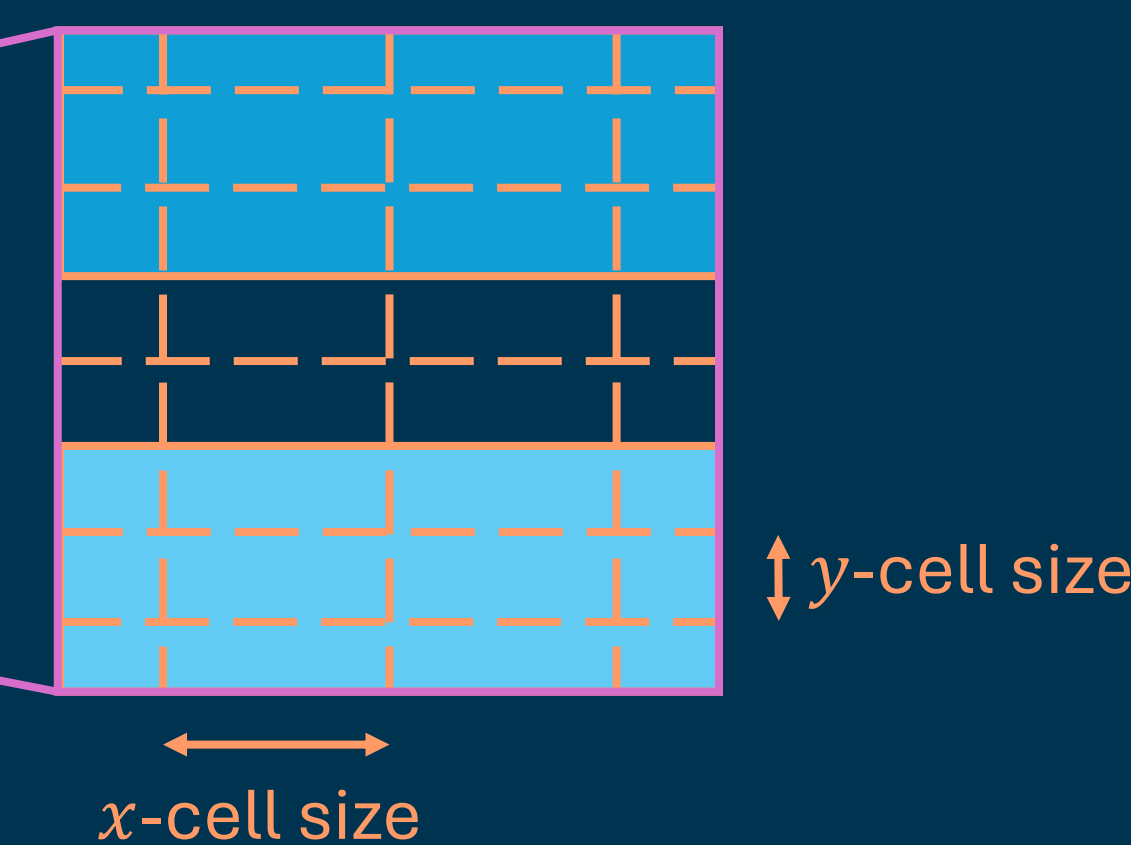
MKID Parameters

The first set of parameters relate to the whole MKID and are controlled through the MKID tab in the interface.

The **size** parameters control the length of the x - and y -axes of the design in micrometres (μm). This is coupled to the maximum sizes of any of the other dimensions of the resonator.

Simulation **cell size** governs the smallest unit cell that will be simulated in Sonnet. This also governs the minimum length and step size of any of the other dimensions of the resonator. These can be different on each axis.

The **kinetic inductance** parameter applies to the whole circuit, but is of particular relevance to the inductor



Inductor

The inductor is constructed of a **Junction** with the Capacitor, a number of **Turns** and an **End** Methods

The inductor tab controls the Breadth and Pitch of the turns, and how many of them are present.

The user sets the Inductor x -size, while its y -size is determined from the pitch and turn count/

